

CHERBURG MAUPERTUS, France

WMO: 070240

Lat: 49.6528N

Lon: 1.4636W

Elev: 459

StdP: 14.45

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.2	32.2	23.1	17.7	33.4	25.5	19.9	34.5	32.3	47.1	29.8	46.4	8.8	160	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	75.2	64.9	71.8	62.8	69.3	61.8	66.1	72.5	64.4	69.6	63.1	67.5	9.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.6	89.6	68.7	62.3	85.5	66.7	61.1	81.9	65.3	31.0	72.5	29.7	69.8	28.8	67.6	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.5	24.5	21.8	DB	26.6	84.9	2.9	4.1	24.5	87.9	22.8	90.3	21.1	92.5	19.0	95.5
			WB	25.2	71.0	3.0	2.3	23.1	72.6	21.3	74.0	19.7	75.3	17.5	77.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	42.9	42.4	45.0	48.8	53.1	58.2	61.7	61.3	59.0	54.4	48.4	44.3	(6)
(7)		DBStd	8.27	5.16	4.90	4.87	4.73	4.41	4.24	3.88	3.34	3.95	4.62	4.52	5.08	(7)
(8)		HDD50	977	223	216	168	75	17	1	0	0	0	13	81	183	(8)
(9)		HDD65	4906	686	634	622	485	370	210	119	125	186	329	499	641	(9)
(10)		CDD50	1584	2	3	11	40	112	246	361	349	270	150	33	7	(10)
(11)		CDD65	35	0	0	0	0	0	5	15	9	5	1	0	0	(11)
(12)		CDH74	174	0	0	0	0	4	34	79	39	15	3	0	0	(12)
(13)		CDH80	31	0	0	0	0	1	6	18	6	0	0	0	0	(13)
(14)	Wind	WSAvg	11.0	13.1	12.4	12.2	10.3	10.5	9.9	9.6	9.4	9.8	10.5	11.5	13.0	(14)
(15)	Precipitation	PrecAvg	35.5	4.0	3.0	2.4	2.4	2.0	1.9	1.9	2.3	2.4	4.0	4.5	4.7	(15)
(16)		PrecMax	45.6	7.2	6.4	6.2	6.1	4.5	4.4	4.1	5.9	4.8	8.5	11.3	10.0	(16)
(17)		PrecMin	25.2	0.7	0.5	0.5	0.3	0.4	0.5	0.6	0.4	0.4	1.1	1.3	0.6	(17)
(18)		PrecStd	5.1	1.9	1.7	1.3	1.5	1.1	1.1	0.9	1.4	1.1	1.9	2.2	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.3	57.3	62.1	69.3	71.9	78.4	82.2	79.2	76.4	71.0	60.9	56.2	(19)
(20)		0.4%	MCWB	52.5	50.2	53.3	58.0	61.4	66.6	68.7	66.8	65.5	62.6	58.3	53.7	(20)
(21)		2%	DB	53.1	52.8	57.2	63.3	67.3	72.3	75.2	72.7	70.6	65.4	58.0	54.6	(21)
(22)		2%	MCWB	51.6	49.7	51.4	54.8	59.2	63.0	64.5	64.2	62.8	60.1	55.4	52.7	(22)
(23)		5%	DB	51.6	50.9	54.0	59.3	63.9	68.8	71.7	69.7	67.5	62.7	56.3	53.0	(23)
(24)		5%	MCWB	50.2	48.8	50.1	52.8	57.4	61.0	63.0	62.7	61.8	58.9	54.0	51.1	(24)
(25)		10%	DB	50.0	49.4	51.8	56.3	60.7	65.8	68.8	67.3	65.1	60.7	54.9	51.6	(25)
(26)		10%	MCWB	48.3	47.5	48.8	51.1	55.6	59.8	61.7	61.4	60.4	57.9	52.7	49.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.5	52.2	54.5	58.7	62.7	68.4	69.5	68.1	66.4	63.5	59.0	54.7	(27)
(28)		0.4%	MCDWB	53.8	54.6	59.2	66.7	68.2	76.3	80.6	76.5	72.6	68.0	60.4	55.4	(28)
(29)		2%	WB	52.0	50.4	52.5	55.8	60.3	64.4	65.7	65.5	64.2	61.1	56.2	53.0	(29)
(30)		2%	MCDWB	52.8	51.8	55.9	62.1	66.1	70.8	72.4	70.4	68.7	63.8	57.6	54.1	(30)
(31)		5%	WB	50.4	49.1	50.8	53.7	58.0	62.1	64.1	63.8	62.7	59.8	54.6	51.7	(31)
(32)		5%	MCDWB	51.4	50.5	53.5	58.3	62.7	67.0	69.7	67.8	66.1	62.0	56.0	52.8	(32)
(33)		10%	WB	48.7	47.6	49.2	51.7	56.2	60.3	62.6	62.5	61.3	58.4	53.0	49.9	(33)
(34)		10%	MCDWB	49.8	49.1	51.5	55.4	60.1	64.8	67.5	66.0	64.2	60.5	54.6	51.3	(34)
(35)	Mean Daily Temperature Range		MDBR	7.2	8.0	9.2	11.6	11.7	12.4	12.7	12.0	11.3	10.1	8.2	7.4	(35)
(36)		5% DB	MCDBR	7.6	9.9	12.9	16.6	17.2	18.0	18.0	16.1	14.4	12.1	9.6	7.8	(36)
(37)			MCWBR	7.9	7.7	8.7	10.3	10.7	10.9	10.0	9.1	8.3	8.3	8.2	7.8	(37)
(38)			MCDBR	7.7	8.5	11.5	15.4	15.3	16.2	16.0	14.2	13.1	10.5	9.1	7.9	(38)
(39)		5% WB	MCWBR	8.4	7.5	8.6	10.0	10.3	10.7	9.7	9.2	8.5	8.1	8.7	8.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.344	0.376	0.389	0.394	0.390	0.384	0.379	0.368	0.357	0.332	0.322	(40)	
(41)		taud	2.420	2.409	2.325	2.313	2.347	2.387	2.405	2.440	2.452	2.466	2.452	2.388	(41)	
(42)		Ebn at Noon	226	252	262	272	274	276	276	272	263	245	223	210	(42)	
(43)		Edh at Noon	21	26	33	37	38	36	35	33	29	24	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	278	509	893	1413	1674	1861	1792	1486	1131	635	339	225	(44)	
(45)		RadStd	24	60	97	120	123	178	116	101	72	52	41	28	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (43 neighbors)	(m) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page